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S11-07

Aim: Write a program for implementing switch case.

Program:

```
import java.util.Scanner;

public class NumberOperations
{
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int number = scanner.nextInt();

        System.out.println("1. Check Prime");
        System.out.println("2. Check Even/Odd");
        System.out.println("3. Check Armstrong Number");
        System.out.println("4. Check Perfect Square");
        System.out.println("5. Check Palindrome");
        System.out.println("6. Calculate Factorial");
        System.out.println("7. Reverse Number");
        System.out.println("8. Calculate Sum of Digits");
        System.out.println("9. Print Triangular Pattern");
        System.out.println("10. Print Fibonacci Series");
        System.out.print("Enter your choice (1-10): ");
        int choice = scanner.nextInt();
```

```
switch (choice)
{
    case 1:
        checkPrime(number);
        break;
    case 2:
        checkEvenOdd(number);
        break;
    case 3:
        checkArmstrong(number);
        break;
    case 4:
        checkPerfectSquare(number);
        break;
    case 5:
        checkPalindrome(number);
        break;
    case 6:
        System.out.println("Factorial of " + number + " is: " + factorial(number));
        break;
    case 7:
        System.out.println("Reverse of " + number + " is: " + reverseNumber(number));
        break;
    case 8:
        System.out.println("Sum of digits of " + number + " is: " + sumOfDigits(number));
        break;
    case 9:
        printTriangularPattern(number);
        break;
```

```

        case 10:
            printFibonacciSeries(number);
            break;
        default:
            System.out.println("Invalid choice. Please choose a number between 1 to 10.");
            break;
    }
}

```

```

public static void checkPrime(int number)
{
    boolean isPrime = true;
    if (number <= 1)
        isPrime = false;
    else
    {
        for (int i = 2; i <= Math.sqrt(number); i++)
        {
            if (number % i == 0)
            {
                isPrime = false;
                break;
            }
        }
    }
}

```

```

    System.out.println(number + " is" + (isPrime ? " a prime number." : " not a prime
number."));
}

```

```
public static void checkEvenOdd(int number)
{
    System.out.println(number + " is" + (number % 2 == 0 ? " an even number." : " an odd
number."));
}
```

```
public static void checkArmstrong(int number)
{
    int originalNumber = number;
    int sum = 0;
    int digits = (int) Math.log10(number) + 1;

    while (number > 0)
    {
        int digit = number % 10;
        sum += Math.pow(digit, digits);
        number /= 10;
    }

    System.out.println(originalNumber + (sum == originalNumber ? " is an Armstrong
number." : " is not an Armstrong number."));
}
```

```
public static void checkPerfectSquare(int number)
{
    int sqrt = (int) Math.sqrt(number);

    System.out.println(number + (sqrt * sqrt == number ? " is a perfect square." : " is not a
perfect square."));
}
```

```
public static void checkPalindrome(int number)
{
    int originalNumber = number;
    int reverse = 0;

    while (number > 0)
    {
        int digit = number % 10;
        reverse = reverse * 10 + digit;
        number /= 10;
    }

    System.out.println(originalNumber + (originalNumber == reverse ? " is a palindrome." : "
is not a palindrome."));
}
```

```
public static int factorial(int number)
{
    if (number == 0)
        return 1;
    else
        return number * factorial(number - 1);
}
```

```
public static int reverseNumber(int number)
{
    int reverse = 0;
    while (number > 0)
    {
        int digit = number % 10;
```

```
        reverse = reverse * 10 + digit;
        number /= 10;
    }
    return reverse;
}
```

```
public static int sumOfDigits(int number)
{
    int sum = 0;
    while (number > 0)
    {
        int digit = number % 10;
        sum += digit;
        number /= 10;
    }
    return sum;
}
```

```
public static void printTriangularPattern(int number)
{
    for (int i = 1; i <= number; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            System.out.print(j + " ");
        }
        System.out.println();
    }
}
```

```
public static void printFibonacciSeries(int number)
{
    int a = 0, b = 1;
    System.out.print("Fibonacci Series: ");
    for (int i = 0; i < number; i++)
    {
        System.out.print(a + " ");
        int sum = a + b;
        a = b;
        b = sum;
    }
    System.out.println();
}
}
```

Output:

Enter a number: 5

1. Check Prime
2. Check Even/Odd
3. Check Armstrong Number
4. Check Perfect Square
5. Check Palindrome
6. Calculate Factorial
7. Reverse Number
8. Calculate Sum of Digits
9. Print Triangular Pattern
10. Print Fibonacci Series

Enter your choice (1-10): 9

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5